

Tips Hindawi Challenge · June–July 2026

# NOVELLM

An AI-Powered Reading Companion

**Author**

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**Organization**

Edrak for AI

**Program**

LLMs Internship

**Model Core**

Mistral-7B · 4-bit



01 / discription

# WHAT IS NOVELLM?

Upload any PDF book and get instant AI grounded insight. Ask questions, generate catch up summaries, explore character networks, quiz yourself, simplify dense passages and discover your next read. All in one unified Gradio dashboard, running locally on GPU with a shareable public link.

Mistral-7B-Instruct-v0.2

LangChain

FAISS Vector Store

Pyvis Graph

PyMuPDF + EasyOCR

BitsAndBytes 4-bit

Gradio

Kaggle GPU

# READING IS HARD. UNDERSTANDING IS HARDER.

## **Context Lost Mid Book**

Hard to recall plot points, character arcs, or key events when returning to a book after a break.

## **Forgetting a Character**

Long books introduce dozens of characters. By chapter 20, readers often can't remember who a minor character is, what role they play, or how they relate to others.

## **Dense Passages**

Complex literary or technical language creates friction — especially for non-native readers or students.

## **No Time to Read It All**

Busy readers need the key points fast not hundreds of pages of dialogue. There's no easy way to get a sharp, spoiler aware summary of exactly what you've missed.

03/ features

# 9 CAPABILITIES IN ONE DASHBOARD

## PDF Upload & Indexing

Full book ingestion with OCR  
fallback for Arabic & scanned text

## Full Book Q&A

RAG-grounded answers from the  
actual text, not model memory

## Catch-Up Summary

Spoiler-free recap up to your  
current chapter via slider

## Character Network

Auto-extracted character cards +  
interactive relationship graph

## Analytics Dashboard

Word count, reading time, chunk  
count, mention frequency chart

## Excerpt Summarizer

Bulleted summaries of any pasted  
excerpt or the whole book

## Quiz Generator

Auto-generated multiple-choice  
comprehension questions

## Explain This

Simplifies dense or difficult  
passages into plain language

## Book Recommender

Mood-based suggestions by vibe,  
pacing, ending type, protagonist  
archetype, length & genre



# HOW IT WORKS

PDF Input  
PyMuPDF parser +  
EasyOCR fallback



Chunking  
Recursive Character  
TextSplitter



Embeddings  
Sentence-transformers  
→ FAISS index



Retrieval  
Semantic similarity  
search



Generation  
Mistral-7B 4-bit  
inference



Dashboard  
Gradio UI + ngrok  
public link

## Quantization

NF4 4-bit quantization via  
BitsAndBytes keeps VRAM  
usage under ~2GB on  
Kaggle T4 GPU

## Custom Output Parser

Extracts structured JSON  
(characters, quizzes,  
recommendations) from raw LLM  
text output reliably

## OCR Fallback

EasyOCR handles Arabic-  
language and scanned PDFs  
that PyMuPDF cannot parse  
digitally

# THE RAG PIPELINE

## 01 Text Extraction

PyMuPDF reads the PDF digitally. If that fails (scanned/Arabic), EasyOCR takes over page-by-page.

## 02 Chunking

RecursiveCharacterTextSplitter splits text into overlapping chunks for semantic coherence.

## 03 Embedding + Indexing

Sentence-transformers encode chunks; FAISS indexes them for fast similarity search.

## 04 Retrieval

Query is embedded; top-k most relevant chunks are retrieved from the FAISS index.

## 05 Generation

Retrieved chunks + user query are passed to Mistral-7B (4-bit) via LangChain pipeline wrapper.

### Key Technical Choices

VRAM Usage (4-bit NF4)



~2 GB vs ~14 GB (full precision)

Retrieval Precision



Grounded in source text

Language Support



English + Arabic (OCR)

### Why Local RAG?

Answers are grounded in the actual book text not hallucinated from model weights. Users get page-accurate, spoiler-aware responses.

# CHARACTER NETWORK ANALYSIS

## 01 LLM Character Extraction

Mistral-7B scans the full text and identifies characters, roles, and relationships

## 02 Custom Output Parser

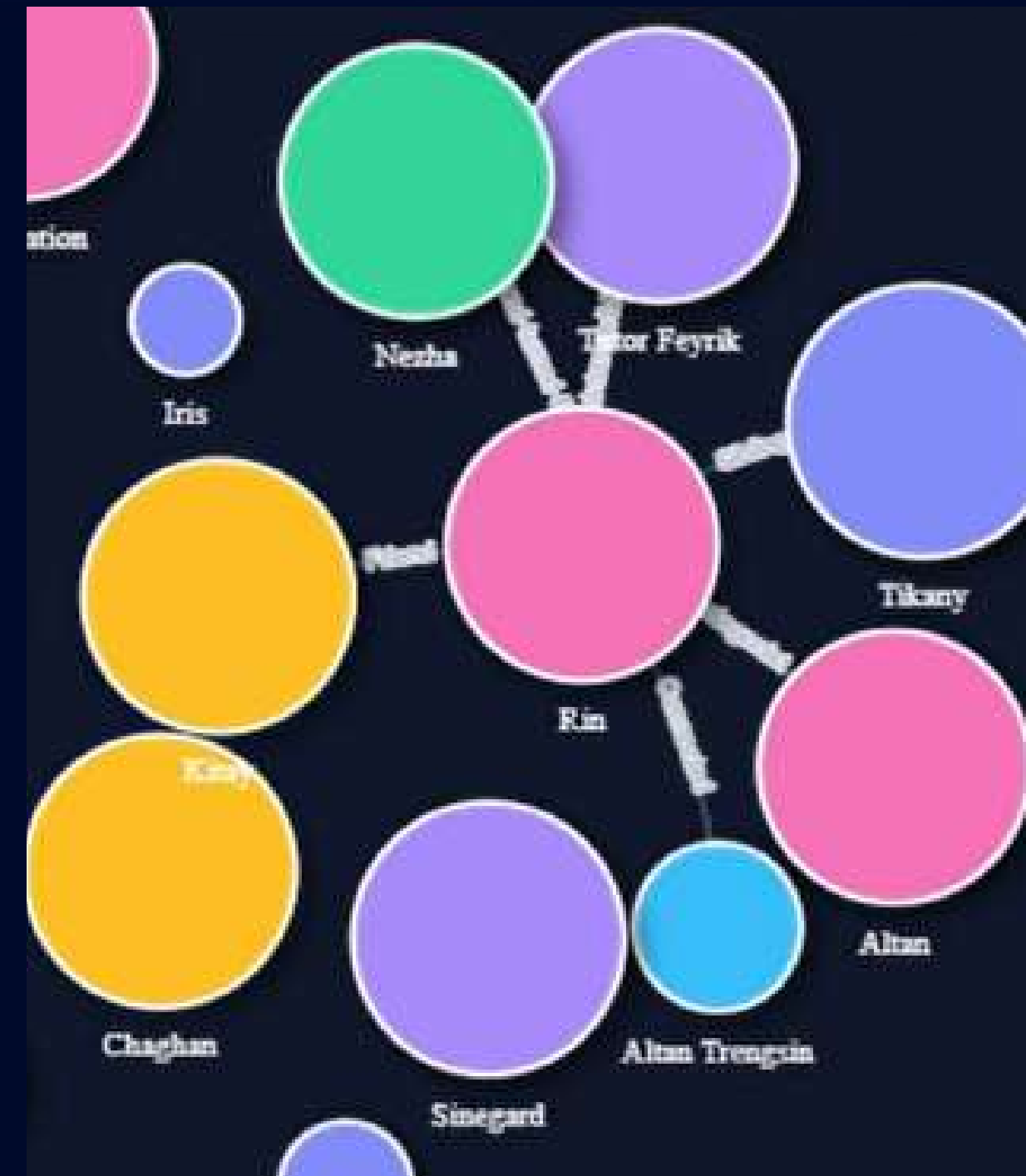
Raw LLM text parsed into structured JSON: name, role, mentions, summary, relationships

## 03 Character Cards

Each character rendered as a clickable card with mention count and role tag

## 04 Pyvis Relationship Graph

Interactive network graph — nodes = characters, edges = relationships, rendered in browser





# LLM CURRICULUM COVERAGE

## Mistral-7B Model

Core LLM for all generation tasks – Q&A, summarization, character extraction, quiz generation, and recommendations

## RAG (Retrieval-Augmented Generation)

FAISS vector store + LangChain retriever ensures answers are grounded in the actual book text

## Summarization

Dedicated excerpt summarizer + spoiler-free catch-up summaries, demonstrating extractive & abstractive summarization

## Quantization (4-bit NF4)

BitsAndBytes NF4 quantization applied to Mistral-7B – reduces VRAM from ~14 GB to ~2 GB on Kaggle T4

## Output Parser (Chaining)

Custom flexible parser reliably converts raw LLM text → structured JSON for characters, quizzes, and recommendations

## Novel Addition: Bilingual + Graph

Arabic OCR support + interactive character relationship graph go beyond curriculum – real-world gaps addressed



# FUTURE IMPROVEMENTS

## ● Mobile App (iOS / Android)

Package the reading companion as a native mobile app for on-the-go use — access your book's AI from anywhere

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## ● Domain Fine-Tuning

Fine-tune Mistral on literary datasets to improve domain knowledge, reduce generic responses, and better understand narrative structure

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## ● More File Formats

Add EPUB, plain text (.txt), and DOCX support beyond the current PDF-only pipeline

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## ● User Accounts & Reading Progress

Persistent reading history synced across devices — pick up exactly where you left off, across books

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## ● Audio / Text-to-Speech

Let users listen to summaries, explanations, and catch-ups — turning the companion into an audiobook assistant

# DEMO

Kaggle Notebook Editor

Novellm

+

→ 8b7eecbb206275d869.gradio.live ☆ Incognito

mail YouTube Maps



## Novellm

Interactive RAG Engine • Dynamic Character Graphs • Smart Explanations

● System Ready

 Reader Hub

 Character Network

 Summaries & Quizzes

 Explain This

 Book Recommender

### 1. Ingest & Index Book

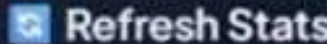
Upload Book (PDF)



Drop File Here  
- or -  
Click to Upload

 Index Book

Analytics Sidebar

 Refresh Stats

No book loaded yet.

Mention Frequency



Indexing Status



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# THANK YOU

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